

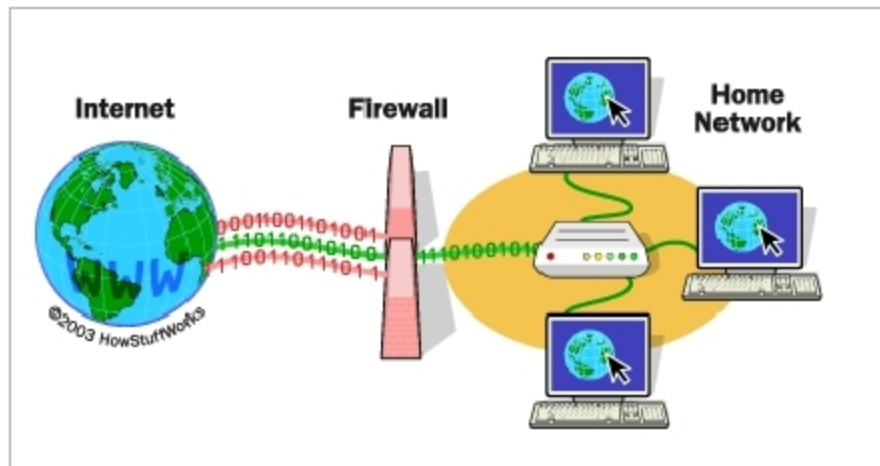


Figure 1: Overview of a firewall

Hardware firewalls have become more advanced these days, and are capable of analysing data on various parameters, including the time of entry into a network, etc.

Software firewalls: For individual home users, the most popular are software firewalls. These can be installed like any other computer software, allowing users to control the functions and choose protection policies. A software firewall protects the computer from outside attempts to gain control over the system and also contends with a wide variety of Trojans, rootkits, viruses, malicious software, worms and even ransomware. Software firewalls are equipped with privacy controls, Web filtering capabilities and many other features. They run in the background and use limited resources.

There are many differences between hardware and software firewalls, but for the best protection of computers and networks, most administrators use both.

Types of software firewalls

The following are the types of software firewalls:

- Packet filtering firewalls
- Circuit level gateways
- Application level gateways
- Stateful multi-layer inspection firewalls

Packet filtering firewalls: These apply a set of rules, and examine each packet to determine whether to forward or drop a packet towards a particular destination. The firewall is configured to filter the packets going in both directions – inbound and outbound. Packet filtering firewalls permit or deny network traffic on the basis of the following information:

- Source IP address and destination IP address
- Protocols like UDP and TCP
- Source and destination ports
- Direction – inbound or outbound
- A physical interface with packet flow

Circuit level gateways: Circuit level gateways work at the session layer of the OSI or Open Systems Interconnection model. They monitor the TCP handshake between the packets to determine if a requested session is legitimate and whether the information passed through

a circuit level gateway, to the Internet, appears to have come from the circuit level gateway. So, for instance, there is no way for a remote computer or a host to determine the internal private IP addresses of the organisation. This technique is termed as network address translation, whereby the private IP addresses originating from the different clients inside the network are all mapped to the public IP address available via ISP and sent to the outside world.

Application level gateway firewalls: These decide whether to drop a packet or send it based on the information about the application. This is done by setting up proxies on a single firewall for different applications. Both the client and server connect to the proxies instead of connecting directly to each other. So, any sort of suspicious data is dropped by proxies and complex protocols like H.323, SIP, etc, can also be handled. Application level firewalls can look in to individual sessions and decide to drop a packet based on information in the application protocol headers or in the application payload.

Stateful multi-layer inspection firewalls: Stateful multi-layer inspection firewalls combine the aspects of all the above types of firewalls. They filter packets at the network layer, transport layer and application layer, allowing the packets to pass through if these layers pass all of them, individually. Some of them allow direct connection between the client and the server, as they rely on algorithms to recognise and process application layer data instead of relying on application specific proxies. These offer more security, performance and transparency to end users.

Linux based firewalls

A Linux based firewall is controlled by a program called iptables, which handles packet filtering. It is an administration program built inside the Linux operating system. It works at the transport layer and protects the system by making routing decisions after filtering the packets, based on information in the IP packet header. Iptables requires root based administrative privileges to operate and can only be executed by the root user. It is used to set up, maintain and inspect packet filters in the Linux kernel. Every table contains user-defined chains, and each chain consists of lists of rules applied to incoming packets. Every rule specifies what to do with a packet that follows the rules. The rules are related to accepting or dropping the packet entering the private network from the outside world.

In order to select a firewall for Linux, there are various concerns with regard to installation, configuration, documentation, features, operation and even support. There are a large number of firewalls available, like IPCop, Shorewall, Monowall, etc. Shorewall is regarded as one of the most powerful firewalls in Linux.